



Nathan Spelts

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SUMMARY

I write embedded C for automotive diagnostic tools. Before that, I spent nearly three years on safety-critical forklift firmware, secure boot, and Rust CAN tooling. Currently pursuing an M.S. in Cybersecurity at Georgia Tech, focused on cyber-physical systems.

WORK EXPERIENCE

Salvo Software, Vancouver, WA

Apr 2026 – Present

- Aftermarket Automotive Engineer
 - Maintain production embedded C firmware for a widely used vehicle diagnostic tool.
 - Implement ECU software update routines for engine and transmission controllers, with a focus on reliability.
 - Develop cloud services that support the tool.

Hyster-Yale Materials Handling, Fairview, OR

Aug 2023 – Apr 2026

- Software Engineer II
 - Wrote safety-critical embedded C firmware for a truck controller, from pallet jacks to container handlers.
 - Improved secure boot and firmware authentication to guard against tampering and unsafe OTA updates.
 - Built a Rust CAN flashing and diagnostics tool (Vector XL, C FFI) used by about 25 people; it cut 3 minutes from every flash and took plain binaries, not the commercial tool's zip and layout file.
 - Developed Lua applications for an in-truck smart display running embedded Linux.

Seasalt.AI, Remote

Jul 2022 – May 2023

- Data Science Intern
 - Reduced word error rate over 40% for Spanish and Indonesian Kaldi ASR models through model and data tuning.
 - Automated scraping and cleaning of thousands of hours of captioned training audio in Python and Bash.

EDUCATION

Georgia Institute of Technology, Remote

Aug 2025 – Present

- Master of Science in Cybersecurity – *Cyber-Physical Systems*
 - Current GPA: 4.00 / 4.00
 - Coursework: secure system design, binary exploitation, network and database security, applied cryptography.

Washington State University, Vancouver, WA

Graduated May 2023

- Bachelor of Science in Computer Science
 - GPA: 3.90 / 4.00

PROJECTS

Raspberry Pi 4 Rust Bootloader

github.com/micro-wizard/raspi4_rust_bootloader

- Bare-metal no_std Rust boot code and kernel for the Raspberry Pi 4, booted straight from the SD card.
 - Wrote AArch64 startup assembly that parks secondary cores, zeroes BSS and sets up the stack for Rust.
 - Drove the VideoCore GPU via its mailbox interface for a double-buffered framebuffer with text rendering.
 - Unit-tested the hardware-facing code on the host against a mocked mailbox, run in GitHub Actions CI.

SKILLS

Programming Languages: C, C++, Rust, Python, C#, Lua, Bash

Embedded & Systems: Secure Boot, Bootloaders, Bare-Metal ARM, Nucleus RTOS, CMSIS-RTOS

Protocols: CAN, UDS, ISO-TP, J1939, CANopen, I2C, SPI, UART

Tools & Technologies: .NET, Git, GitHub Actions, Docker, Linux, GDB, QEMU, Vector XL